Check for updates

Received: 25 December 2023 | Revised: 26 July 2024 | Accepted: 21 August 2024

IET Communications

DOI: 10.1049/cmu2.12830

The Institution of Engineering and Technology WILEY

REVIEW

Physical layer security in satellite communication: State-of-the-art and open problems

Nora Abdelsalam[1] | Saif Al-Kuwari[1] | Aiman Erbad[2]

[1]Division of Information and Computing Technology, College of Science and Engineering, Hamad Bin Khalifa University, Doha, Qatar

[2]College of Engineering, Qatar University, Doha, Qatar

Correspondence
Saif Al-Kuwari, Division of Information and Computing Technology, College of Science and Engineering, Hamad Bin Khalifa University, Doha, Qatar.
Email: smalkuwari@hbku.edu.qa

Funding information
Qatar National Research Fund, Grant/Award Numbers: GSRA8-L-2-0521-21046, AICC03-0530-200033

Abstract

Satellite communications have emerged as a promising extension of terrestrial networks in future 6G network research due to their extensive coverage in remote areas and their ability to support the increasing traffic rate and heterogeneous networks. Like other wireless communication technologies, satellite signals are transmitted in a shared medium, making them vulnerable to attacks such as eavesdropping, jamming, and spoofing. A good candidate to overcome these issues is physical layer security (PLS), which utilizes physical layer characteristics to provide security, mainly due to its suitability for resource-limited devices such as satellites and IoT devices. This paper provides a comprehensive and up-to-date review of PLS solutions to secure satellite communication. Main satellite applications are classified into five domains: satellite-terrestrial, satellite-based IoT, satellite navigation systems, FSO-based, and inter-satellite. In each domain, how PLS can improve the overall security of the system, preserve desirable security properties, and resist widespread attacks are discussed and investigated. Finally, some gaps in the related literature are highlight and open research problems, including uplink secrecy techniques, smart threat models, authentication and integrity techniques, PLS for inter-satellite links, and machine learning-based PLS, are discussed.

1 | INTRODUCTION

The demand for high bandwidth and reliable communication is rapidly increasing due to the increased number of connected devices and advanced technologies. Current terrestrial networks have limited resources and spectrum, which cannot meet the requirements of next-generation applications and the increasing number of Internet of Things (IoT) devices. In addition, current cellular networks cannot cover all geographical areas, especially remote locations with low populations, because network installation in these areas is expensive and challenging [1]. As a result, satellite networks have gained increasing interest in recent years as a promising solution to the limitations of terrestrial networks.

Satellite communication systems cover large areas due to their broadcast nature, providing a large-scale footprint service and flexible connectivity [1, 2]. However, satellite links suffer from a significant round trip time that increases latency and decreases service quality. Therefore, using satellite links alone will not meet the expected network requirements. Leveraging the benefits of terrestrial and satellite networks, the researchers start designing new network architecture to achieve the quality of service required with ubiquitous connectivity. Furthermore, there is a significant interest in satellite constellation projects such as Starlink, which SpaceX initiated in 2015 to provide high-speed internet connection worldwide using LEO satellites [3].

Signals sent from satellites are transmitted on a shared medium and can be received by all surrounding nodes, both legitimate and malicious. Hence, satellite communications are vulnerable to both passive and active attacks. In passive attacks, such as eavesdropping, the adversary observes the data transmitted through the channel, attempts to retrieve confidential information, and causes information leakage [4]. On the other hand, in active attacks, such as spoofing, jamming, and data manipulation, the adversary tampers with the traffic during its transmission. For example, the attacker can pretend to be a legitimate satellite or station and obtain illegal access to sensitive information. These attacks can cause ripple effects to satellite networks and enormous damage, such as distraction, denial of service, and economic losses.

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There are two approaches to secure such communication networks: the upper layer approach and the physical layer approach. The upper layer approach is based on classical cryptographic techniques, including symmetric cryptography (using a secret key shared by legitimate users) and public key cryptography, and it proved stable against many widespread attacks. However, encryption and decryption are expensive processes that require high computational power, which is not usually available for satellites and IoT devices [5]. Furthermore, satellites usually find key management and distribution problematic because they require complex architectures and protocols. Moreover, cryptographic techniques assume that the adversary lacks the computational capabilities to break its schemes, such as RSA, making it vulnerable once fault-tolerant quantum computers are available [6].

On the other hand, physical layer security (PLS) techniques are based on the wiretap model introduced by Wyner [7], in which information-theoretic security can be achieved without pre-shared keys. This model has two channels: the main channel between legitimate users (transmitter and receiver) and the wiretap channel between the legitimate transmitter and the eavesdropper. PLS aims to reduce the quality of the wiretap channel compared to the main channel to maximize the mutual information difference between the two channels. PLS uses wireless channel characteristics such as reciprocity and spatial and temporal variations to encode the information exchanged.

1.1 | Existing surveys

PLS is a lightweight approach applicable for securing many wireless communication systems, such as satellites and IoT devices. Many works in the literature have recently focused on designing PLS schemes to secure terrestrial and satellite networks with different architectures.

Some existing surveys mainly discuss satellite evolution from multiple perspectives. Surveys such as [8] and [1] review the convergence of terrestrial and satellite network integration. The authors of [8] identify the critical building blocks and the applicability of current networks for convergence from architecture and performance aspects. Similarly, [1] is a comprehensive survey that discusses the motivation and requirements for the intersection of satellite and terrestrial networks and the key enablers.

Another type of survey focuses on the physical layer techniques and optimization approaches. The authors in [9] discussed the fundamentals of PLS, including attack and wiretap channel models. Similarly, the authors in [10] divide PLS types against passive attacks into signal-to-interference-plus-noise ratio-based and complexity-based approaches (key-based). Likewise, the authors in [6] focus on SNIR approaches and review works related to secure beamforming and antenna node selection. The authors in [5] provided a broader view of PLS techniques and reviewed the three main security properties: node authentication, message integrity, and confidentiality.

Some surveys cover the security aspect of satellite communication [2, 11, 12]. The authors in [12] discuss the vulnerabilities and security challenges of LEO satellite systems, including passive eavesdropping, active attacks, and interference with possible countermeasures. On the other hand, the authors in [11] focus on security enhancement techniques, dividing them into physical layer security mechanisms and cryptography. They consider the PLS schemes that ensure the confidentiality, authentication, and availability of satellite communications. Lastly, [2] studies the state-of-the-art PLS implementations in three satellite network architectures: mobile terrestrial satellite networks, hybrid satellite and terrestrial relay networks, and integrated networks. A comparison of existing surveys is available in Table 1. In our comparison, we focus specifically on PLS in satellite domains, which multiple surveys in the literature do not cover.

1.2 | Contribution

Extending terrestrial coverage is generally the main application of satellites. However, satellites can help realize many other essential and practical applications. For example, satellite systems are vital in navigation applications, which continuously provide positions and time information for any location on Earth. Global navigation satellite systems are a growing field of development, from the global positioning system (GPS) project in 1973 to regional systems projects planned to be launched in 2023 [13]. Similarly, satellite-based IoT is a promising domain that leverages satellite coverage to connect IoT devices distributed in prominent geographical locations and harsh environments, where mission-critical IoT applications such as rescue operations or disaster recovery can finally be efficiently realized [14]. Furthermore, satellites are prominent in optical communication, where free-space optical (FSO) networks are extended to cover increasingly larger areas via satellites. Eventually, satellite networks will grow in size and maturity, necessitating establishing and maintaining communication links between the satellites (inter-satellite communication), inevitably leading to even more exciting applications.

Existing surveys provide reasonable coverage of the use of PLS in several satellite networks. However, the literature lacks a comprehensive survey covering multiple satellite-based domains, which we attempt to address in this survey.

In this paper, we classify satellite applications into five main application domains and analyse PLS in each field. These domains are 1) terrestrial and satellite networks, 2) navigation satellite systems, 3) satellite-based IoT, 4) FSO-based systems, and 5) inter-satellite communications. There are many other applications for satellites, such as those related to military and research, but, to the best of our knowledge and our literature analysis, we found that these five areas are the main domains where satellites play a significant role. Therefore, it is critical to analyse the security threats for each of these domains and study how PLS can provide the necessary protection, utilizing the unique characteristics of each domain.

The contributions of this survey can be summarized as follows.

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TABLE 1 Comparison between existing surveys and this survey.

Survey	Year	IOT networks	GNSS satellites	Satellite-terrestrial networks	FSO satellites	Inter-satellites
[8]	2016	✗	✗	✗	✗	✗
[9]	2016	✗	✗	✗	✗	✗
[10]	2018	✓	✗	✗	✗	✗
[6]	2018	✗	✗	✗	✗	✗
[1]	2019	✗	✗	✗	✗	✗
[2]	2019	✓	✗	✓	✗	✗
[11]	2021	✗	✓	✓	✗	✗
[12]	2022	✗	✗	✗	✗	✗
Our survey	2023	✓	✓	✓	✓	✓

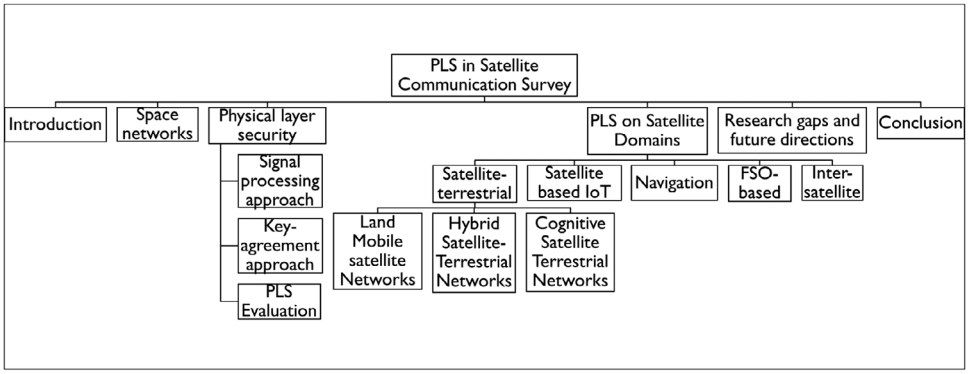


FIGURE 1 Organization of this survey paper.

- We classify satellite networks as follows: satellite-terrestrial, satellite-based IoT, navigation, FSO-based, and inter-satellite, as shown in Figure 1. We highlight the primary usage for each domain and discuss their security concerns and vulnerabilities.
- We review the state-of-the-art works of PLS techniques for each satellite domain, investigate the design of each use case, and focus on its security analysis. We extensively compare all available solutions for each area and discuss their goals and results.
- We identify future trends and directions for PLS development in satellite networks.

1.3 | Organization

The rest of this paper is organized as follows: Section 2 provides the necessary background for satellite and space networks while Section 3 provides background for PLS. In Section 4, we discuss the five domains of satellite networks. For each domain, we describe the architecture of the satellite network and its importance, review recent PLS schemes, and compare them. Finally, we conclude this paper in Section 5 by highlighting open challenges and future directions of PLS in satellite networks. Figure 1 provides a visualization of the structure of this survey.

2 | SPACE NETWORKS

Researchers propose several integration architectures to leverage the benefits of both satellites and terrestrial networks. A generic architecture illustrated in Figure 2 extends the current system to achieve global coverage by dividing terrestrial-satellite networks into three main parts: terrestrial networks, air-based networks, and satellite networks. This architecture is extracted from several research papers such as: [1, 15], and [16] where the data is transmitted using different links between these three layers utilizing their distributed resources to provide global connectivity and ubiquitous access. The Terrestrial networks layer includes conventional wired and wireless communication networks, satellite ground stations, and all user terminals on the Earth's surface. The next level is the air-based networks, including high-altitude platform stations (HAPS), unmanned autonomous vehicles (UAVs), and planes. The last level is the satellite networks, which include three types of satellites: low earth orbit (LEO), medium earth orbit (MEO), and geostationary earth orbit (GEO).

Different satellite networks operate at different orbital heights, resulting in various delays and coverage areas. GEO satellites operate at an orbit height of 36,000 km, covering a large area of the earth, which increases their availability, but introduces latency due to more extended round trip. Hence,

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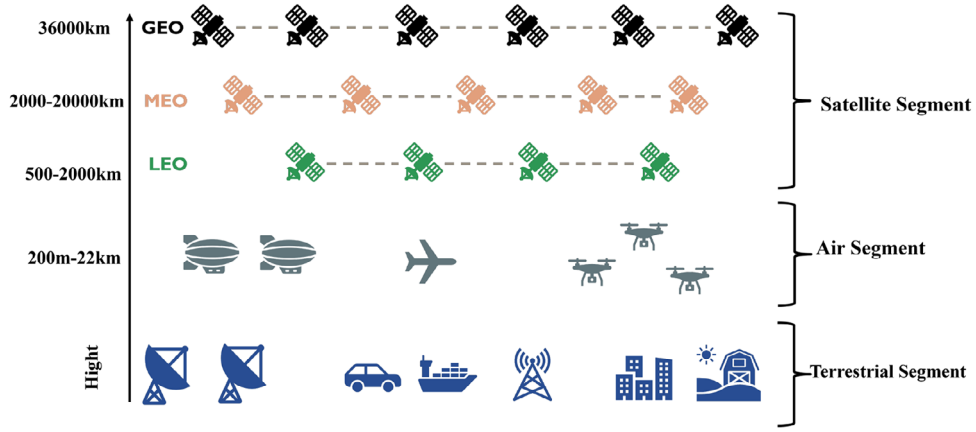


FIGURE 2 Satellite and terrestrial integration architecture.

GEO satellites work as the space backbone network and are responsible for network management. MEO satellites operate in lower orbits at a height of 2000–20,000 km, which decreases the latency but increases the number of satellites needed to cover the Earth. LEO satellites are the closest to Earth, operating on 500-2000 km, with decreased latency, path loss, and attenuation. Hence, LEO satellites are considered access points to space networks with inter-satellite links.

With their communication links, all these types of satellites are vulnerable to several attacks. Like conventional adversaries, space adversaries can be either passive (attempting to compromise the confidentiality of the communication) or active (attempting to spoof the communication by tampering with the traffic). Furthermore, adversaries can target the service availability of satellite systems by jamming or compromising them. Therefore, various security concerns should be considered for each satellite network design and architecture.

3 | PHYSICAL LAYER SECURITY (PLS)

PLS was first introduced by Wyner [7], who proposed a wiretap channel model. This model secures the channels of legitimate users from unauthorized eavesdropping or signal interception. It aims to utilize the wireless channel noise and randomness to outperform the quality of the eavesdropper's channel (wiretap channel). When the wiretap channel is degraded, the mutual information difference between the two channels is maximized, preventing the eavesdropper from being able to decode the transmitted signals. Mutual information is the amount of information the receiver obtains from the original message from the received one. Mutual information measures the reliability of the link because it quantifies the amount of information sent correctly to the receiver. Therefore, PLS aims to maximize the mutual information of the legitimate link (increase reliability) while minimizing it on the wiretap link, which increases randomness and reduces the amount of information received by the eavesdropper [17].

PLS techniques are unconditionally secure (assuming a threat model where adversaries have access to unlimited resources), which makes them quantum-resistant by nature. Furthermore, PLS does not require pre-shared keys between legitimate users, which solves the current key management and distribution difficulties. Finally, PLS is lightweight, making it an ideal choice for IoT applications.

In the following section, we discuss some dominant PLS approaches in the literature.

3.1 | Signal processing approach

The PLS signal processing approach mainly increases the main channel's SINR (signal-to-interference and noise ratio) and decreases it for the wiretap channel. As a result, the eavesdropper will have a more noisy channel (degraded) that prevents the eavesdropper from accessing the transmitted data. This keyless approach can be based on channel adaptation or inserting artificial signals to degrade the wiretap channel. An example of enabling technology is the use of multiple antennas. Multiple input multiple output (MIMO) is a multiantenna technology in which the transmitter and receiver have several antennas for communication. These antennas help increase the transmission rate and improve signal strength. MIMO technologies have recently attracted increasing attention in satellite communications [18, 19]. Examples of PLS techniques based on MIMO are listed below:

- Beamforming: a technique that focuses the signal strength in one direction, creating a beam of signals using multiple antennas. Beamforming creates a directive transmission by adjusting the weight of each physical antenna. This method is used to direct the signal towards the legitimate user to maximize his SINR while downgrading the adversary's (as the latter only receives a negligible amount of leaked signal). Figure 3a illustrates how beamforming varies the signal strength between the two channels.
- Zero forcing (ZF): a method that divides the wireless communication channel for multiuser into several independent channels and cancels the interference for these channels. Zero forcing precoding can increase secrecy capacity by

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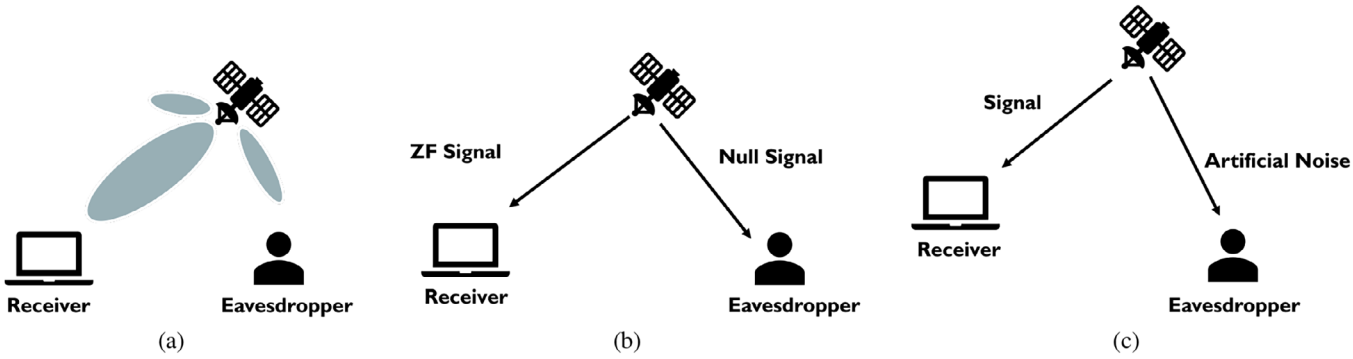


FIGURE 3 Techniques for PLS based on signal processing (a) beamforming technique that directs the signal towards the legitimate user, (b) zero-forcing technique sends messages to cancel interference orthogonal to the eavesdropper's, (c) artificial noise technique sends interference signal to the eavesdropper side.

sending the message to cancel interference orthogonal to the eavesdropper's channel, resulting in a null signal on the eavesdropper's side. Therefore, this technique creates the null signal based on the eavesdropper's channel state information (CSI). Figure 3b illustrates how the ZF signal is orthogonal to the eavesdropper.

- Artificial noise (AN): a physical layer jamming technique where additive white Gaussian noise (AWGN) is designed to align with legitimate users' null space and only affect the eavesdropper. AN is an interference signal that increases the noise in the eavesdropper channel to protect transmitted data. The transmitter will divide the transmission power to transmit the message and send artificial noise to the eavesdropper. The transmitter needs to know the CSI of the eavesdropper to send this artificial noise. This noise is added orthogonal/perpendicular to the data vector means in the null space of the receiver. Therefore, it is designed to be cancelled on the legitimate user side and only affects the illegitimate user. Figure 3c illustrates how artificial noise can be created.

Alternatively, the transmitter can use relay devices to amplify the signal on its way to the receiver [20]. It can increase the secrecy capacity through different cooperative strategies/communication with the transmitter.

3.2 | Physical layer authentication

Physical layer authentication (PLA) is a type of PLS that focuses on identifying devices and granting access to legitimate users only. PLA identifies users based on device fingerprinting by extracting unique features from their signal transmission and using these features to authenticate users in the future. Furthermore, PLA utilizes the characteristics of the physical layer channel of the legitimate user, such as multipath and device movement, for channel-based authentication. Another type of PLA is the active scheme or key-based PLA, which authenticates the transmitter via a shared key with the receiver. This key creates a tag embedded in the message and is used by the receiver to authenticate the sender from the physical layer coding phase without upper-layer processing [21].

3.3 | Key-agreement approach

Key agreement techniques leverage the randomness of the channel between legitimate users to generate secret keys. In this method, there is no assumption that eavesdroppers will have a degraded channel or less SNIR. It works on systems where transmitter and receiver experience theoretically identical wireless channel states while it is different for eavesdroppers, which should be located half wavelength away [10]. In these systems, users do not exchange explicit messages about the channel; they use only pilot messages, which the eavesdropper can use to estimate the channel and recover the key. This method utilizes channel state information, received signal strength (RSS), or phase shift between the legitimate user to extract randomness, which is then quantized into random bits. Next, a reconciliation process is used to remove erroneous bits. Finally, the generated key's randomness is improved using privacy amplification, which removes any correlation between bits. This method is used to achieve confidentiality and authentication via the physical layer.

3.4 | PLS evaluation

Several security metrics for measuring the effectiveness of PLS techniques exist. In this section, we summarize the most commonly used security metrics with their definitions and expressions.

3.4.1 | Secrecy capacity

Secrecy capacity is the highest communication rate that guarantees the reliability of the legitimate link with perfect security, which means that the mutual information for the eavesdropper channel is zero [22]. Hence, it is the difference between the main channel capacity and the wiretap channel capacity. Secrecy capacity can be expressed as:

$$C_s = \max[I(X, Y_B) - I(X, Y_E)] \quad (1)$$

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where I is the mutual information, X is the message sent, and Y_B and Y_E are the messages received by the legitimate user and the eavesdropper, respectively.

3.4.2 | Achievable secrecy rate

The achievable secrecy rate is the difference between the data rate achieved by the main channel and the wiretap channel with the Gaussian codebook. It can be expressed as:

$$R_s = [R_B - R_E]^+ \tag{2}$$

where $[]^+$ means $\max(x, 0)$ and R_B, R_E are the legitimate and eavesdropper link rates, respectively. The achievable secrecy rate is the lower bound of the secrecy capacity, which is used for more computationally affordable calculations since the difference in the secrecy capacity is a non-convex optimization problem [23].

3.4.3 | Secrecy outage probability

Due to variations in channel state and quality, the secrecy rate can be affected over time. If the secrecy rate drops below a specific target rate, the secrecy outage occurs, indicating that the security goal cannot be achieved. The secrecy outage probability can be measured as

$$\text{SOP} = Pr(C_R < R_0) \tag{3}$$

where C_R is the secrecy capacity and R_0 is the target rate [24]. For some SOPs, it is practical to know the likelihood of a secrecy outage to gain more insight about the security level.

3.4.4 | Secrecy energy efficiency

In power-constrained systems, it is essential to consider energy consumption in the design of the security scheme. Secrecy energy efficiency measures the energy consumption of all nodes with the security performance [25]. Specifically, it can be measured as

$$\text{SEE} = R_s / E_{\text{total}} \tag{4}$$

where R_s is the secure bits, and E_{total} is the total energy the system uses. Therefore, SEE measures the number of secure bits transmitted in the energy unit.

3.4.5 | Security gap

The security gap is a practical measure of security performance introduced in [26]. It is based on the bit error rate at the legitimate receiver and eavesdropper and can be expressed as:

$$\text{SG} = \text{SNR}(B)_{\min} - \text{SNR}(E)_{\max} \tag{5}$$

This equation denotes the difference between the minimum signal-to-noise ratio the receiver needs to decode the signal correctly and the maximum signal-to-noise ratio for the eavesdropper, which ensures a specific error rate level. The gap between these ratios indicates the quality advantage that legitimate users should have to guarantee the security requirement [17].

4 | PLS ON SATELLITE COMMUNICATION

In this survey, we classify satellite networks into five domains: hybrid, IOT, navigation, FSO, and inter-satellite. We review each of these domains in the following subsections.

4.1 | Domain 1: satellite-terrestrial networks

Extending terrestrial networks is the primary goal of satellite network integration to overcome current network limitations of coverage and spectrum. Specifically, taking advantage of the large footprint of satellites that integrate satellite and terrestrial networks supports connecting rural and off-shore areas that the existing network infrastructure does not cover. Furthermore, the currently congested spectrum poses severe challenges to high-bandwidth applications. Hence, spectrum sharing between satellite and terrestrial networks allows efficient spectrum use and provides reliable communication. However, several challenges must be solved to achieve this integration, including performance, resources, and security. Integration designs can be categorized into land mobile systems, hybrid, and cognitive networks. In the following subsections, we will review each design architecture's main security concerns and relevant PLS works.

4.1.1 | Land mobile satellite networks

Land mobile satellite (LMS) networks consist of satellites directly connected to ground stations using radio frequency signals. Satellites can be spot beams or multibeams, where the difference is the area of coverage and spectrum reuse. Specifically, multibeam allows coverage of multiple users using a single beam and sends the information to various spots on the ground with frequency reuse for each beam. Figure 4 shows the architecture of LMS with spot beam and multibeam satellites. As the figure shows, the coverage of satellite signals allows illegitimate users to receive the signal, which threatens the confidentiality and authenticity of the link. Hence, several works focus on the security issues of LMS and study PLS performance in such systems.

Ref. [27] consider the security performance of Non-geostationary satellites with different orbit models. Non-geostationary satellites are not in a fixed position for Earth, and they are moving around it in multiple orbits, such as LEO and MEO. They utilize satellite movement to evaluate the security of the system in different positions and

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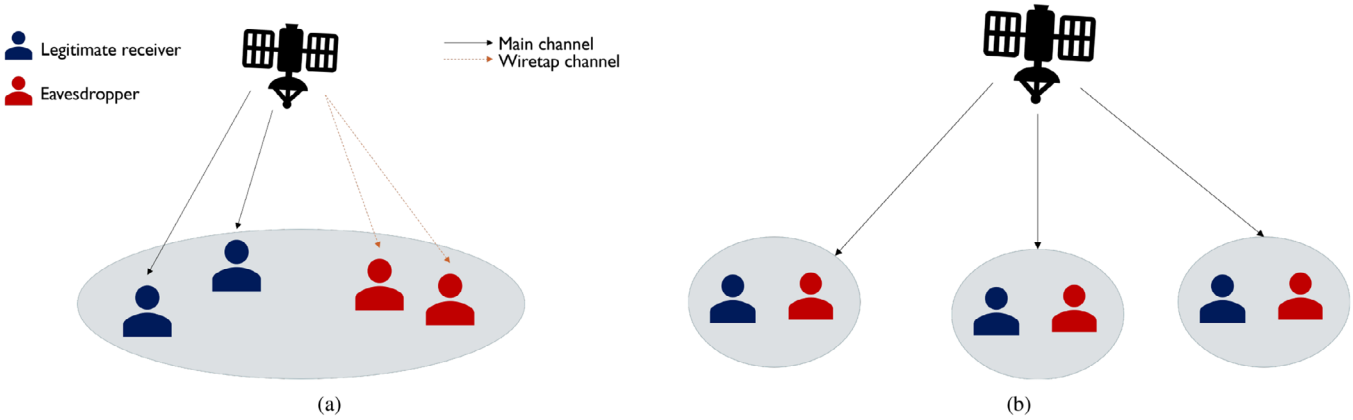


FIGURE 4 Land mobile satellite with multiple users and eavesdroppers (a) one spot beam satellite, (b) multi-beam satellite.

atmospheric conditions, including rain attenuation. They derive the closed-form expression of secrecy capacity and secrecy outage probability and provide a Monte Carlo simulation.

The authors of [28] propose a wiretap channel model with a finite length regime. Their model includes a coding method that creates wiretap code using linear error-correcting codes such as polar codes and randomized hash functions. They consider RF satellite uplink and measure the secrecy capacity for the proposed scheme. Furthermore, their method identifies the spatial regions where secrecy can be granted. Their analysis assumes that the receiver satellite is in GEO or MEO orbits and the eavesdropper is in LEO, MEO orbits, or UAV at different heights.

Subsequently, the authors of [29] and [30] consider an LMS system with multiple users cooperating to receive the message, and multiple eavesdroppers are trying to intercept data sent to legitimate users. They obtain the closed-form expressions for the non-zero probability of secrecy capacity, the secrecy outage probability (SOP), and the average secrecy capacity (ASC). Ref. [29] extends the eavesdropping scenario to two scenes: colliding and non-colliding, where eavesdroppers collaborate together to wiretap the channel in the colliding case, and the best eavesdropper will be selected to wiretap the channel in the non-colliding scenario. Analysis in [30] considers imperfect channel estimation to study the effect of various fading strategies on secrecy performance.

A recent attempt to leverage the advancements in MIMO technology in satellite PLS was conducted in [31], where they used a multi-beam satellite with a single feed-per-beam (SFPB) antenna and full-frequency-reuse (FFR). This architecture allows the satellite to send multiple beams to the earth's surface with the same spectrum and polarization for all beams. They consider two designs for the antenna reflector: one parabolic reflector and multiple reflectors. They aim to maximize the secrecy capacity of each user against its eavesdropper, constrained by the power consumption for each beam. Therefore, they formulate the optimization problem and solve it using convex approximation. Furthermore, the authors introduce artificial noise into the multiple reflector case to increase secrecy when the number of eavesdroppers exceeds the number of beams.

Giving more focus to authenticating satellite-to-ground links, the authors of [32] proposed a physical layer authentication mechanism to authenticate iridium satellites from ground stations. Iridium is a constellation of 66 LEO satellites for voice and data transmission, and dedicated mobile satellite devices or integrated transceivers can receive its signal. The proposed scheme depends on the hardware impairments of the satellites, making their IQ samples different and can be used as a fingerprint. They collect a dataset of IQ samples for each satellite in the formation and process them to create greyscale images. They authenticate satellites in two scenarios: 1) multiclass classification, where each satellite represents one class, and in this case, they use deep CNN for classification; 2) binary classification, where they authenticate only one satellite against others, and here autoencoders are used. Their results showed that artificial intelligence algorithms could authenticate satellites based on their IQ samples.

Similarly, [33] proposes a physical layer authentication method using the Doppler frequency shift of the satellite. Specifically, their approach focuses on authenticating the system information signalling (SIS) messages sent by satellite to allow the user equipment to request random access. Since illegitimate users also receive these messages, they can spoof the SIS messages by sending similar messages to users, which will cause connection failure. The authentication scheme allows the user terminal to estimate the Doppler shift of the received message (current channel) and compare it to the Doppler shift calculated using satellite ephemeris and position. If both calculations are equivalent, then the satellite signal is authenticated. Their scheme outperforms the CSI-based authentication scheme [9] due to the channel correlation drop.

Table 2 summarizes all discussed PLS works related to the LMS system. We compare all works against their security goal, satellite and link types, and their used PLS technique and measurement. The comparison shows that most works focused on confidentiality requirements and assume passive eavesdroppers that are not actively affecting the main channel. Moreover, all

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TABLE 2 Comparison between existing PLS works for land mobile satellite communication.

Work	Security goal	Satellite	Link	Antenna	PLS technique	PLS feature	Security measure
[27] (2018)	Confidentiality	Non-geo orbit	Downlink	Single antenna	Information theory	Orbiting models	Security capacity, secrecy outage probability
[28] (2019)	Confidentiality	RF satellite	Uplink	Rotational symmetry antenna	Wiretap coding	Eavesdropper noise power	security Capacity
[29] (2019)	Confidentiality	Spot beam	Downlink	Single antenna	Information theory	User cooperation	Security capacity, secrecy outage probability, average secrecy capacity
[30] (2019)	Confidentiality	Spot beam	Downlink	Single antenna	Information theory	Channel estimation errors	Security capacity, secrecy outage probability, Average secrecy capacity
[33] (2021)	Authentication	LEO satellite	Downlink	—	Physical layer authentication	Doppler frequency shift	False alarm rate miss detection rate(FAR)
[31] (2021)	Confidentiality	Multibeam	Downlink	multiple-reflector antenna	MIMO precoding - artificial noise	Antenna spatial degree of freedom	Minimum secrecy capacity
[34] (2023)	Authentication	LEO satellite	Downlink	IRIDIUM Beam Active Antenna model RST740	hardware fingerprinting	IQ samples	Classification accuracy

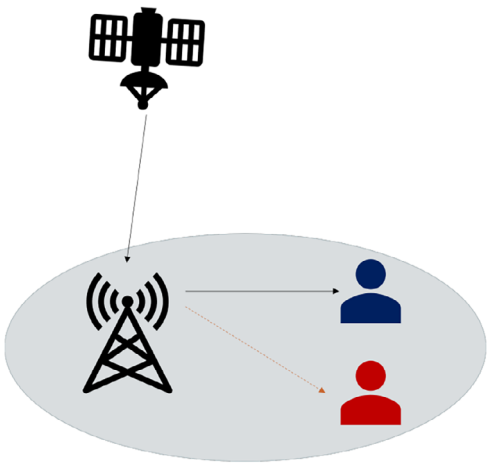


FIGURE 5 Relay-based satellite-terrestrial system.

techniques focus on downlink security from satellite to ground without considering uplink security despite their importance to the satellite networks.

4.1.2 | Hybrid satellite-terrestrial networks

In hybrid satellite-terrestrial networks (HSTNs), the satellite transmits the data to the ground destination with the help of a middle relay, as shown in Figure 5. This relay can be terrestrial stations or air-space devices such as UAVs or HAPS. The relay technology helps to enhance the network performance and security for the ground users [35]. Single or multiple relays can be used in the network to overcome any rain or obstacle stations and increase the confidentiality of the communication. The most adopted relay techniques to increase performance and security rates are discussed below.

- Amplify and forward (AF): This method can be used when the transmitter has limited power to send the message. The relay cooperates with the transmitter by amplifying the transmitter's signal (message) and sending it to the receiver without decoding it. As a result, the transmission rate increases and the receiver noise and interference decrease [22].
- Decode and forward (DF): In this type, the relay cooperates with the transmitter by decoding the received message, re-encoding it, and sending it to the receiver. DF relay location affects the secrecy of the link; the closer the transmitter and the relay are, the higher the secrecy of the link [36].
- Noise and forward (NF): In this type, the relay has two channels. One sends the message to the receiver, and the other sends artificial noise to the eavesdropper's channel. In some cases, this type is used only to send AN to confuse eavesdroppers and increase secrecy capacity.

HSTN is considered a practical design for terrestrial and satellite integration, encouraging researchers to study its performance and security.

The authors in [37] proposed a joint scheme of relay selection and user scheduling and studied its security capacity with multiple users, relays and eavesdroppers. Their scheme depends on measuring the signal-to-noise ratio (SNR) on both links from

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the satellite to the relays and from the relay to the users to choose the best link. The authors studied their scheme under two scenarios. In the first scenario, they assume that eavesdroppers can cooperate to access legitimate information. In contrast, in the second scenario, the eavesdropper with the best signal-to-noise ratio will wiretap the channel. Furthermore, the authors in [38] propose two models of relay selection, single-hop relay, and multihop relay, and compare them with the baseline round-robin scheduling. The single relay scheme chooses the best relay based on the link's instantaneous capacity between the relay and the destination since it assumes a passive eavesdropper with unknown channel information. However, in the multi-relay scheme, multiple relays transmit the data simultaneously with a weighted version of the decoded message. Their schemes outperform the round-robin method. Similarly,[39] investigates the security and reliability of two relay selection schemes in an HSTN system with artificial noise and compares them to the round-robin baseline under different shadowing conditions. They derive the closed-form expressions of the outage probability and intercept probability and present a numerical analysis of the two schemes. Their results showed that increasing the number of relays will increase the reliability of the two schemes; however, it reduces the security where intercept probability increases.

Ref. [40] proposes a PLS framework for downlink HSTN using two relay techniques: amplify and forward, or decode and forward with multiple relays and users. Here, the satellite has multiple antennas where it transmits its signal to all relays, and only the selected relay will amplify or decode the signal and forward it to the user. They assume that eavesdroppers can only wiretap the communication between the relay and users (terrestrial link) with two scenarios of eavesdropping, colluding, and non-colluding. The relay selection mechanism depends on the signal-to-noise ratio between the user and eavesdropper. Their analysis showed that secrecy outage probability is worse in the colluding scenario due to the cooperation between eavesdroppers. Moreover, increasing the number of users enhances the security performance due to the increased diversity gain.

The authors in [41] added power allocation as a constraint to the relay selection scheme to minimize the probability of secrecy outage in HSTN using instantaneous and statistical CSI. In this paper, the relay used is near ground relays, such as airplanes or HAPS, and it transmits an interference signal to downgrade the channel for the eavesdropper with no effect on the quality of the legitimate user channel. Since instantaneous CSI cannot be obtained in satellite communication, they propose a relay selection scheme based on statistical CSI to decrease the signal overhead and delay caused by continuous relay handover. Besides, statistical CSI is used for satellite and relay communications power allocation to minimize the system's energy consumption while enhancing the secrecy level. Similarly, the authors in [42] use the known and unknown CSI conditions for the wiretap channel to propose two relay-user pairing schemes. The optimal relay user pairing scheme assumes complete knowledge of the eavesdropper channel state information. Hence, the relay that can maximize the secrecy rate will be selected. However, the suboptimal method assumes that the full CSI of the eavesdropper is unknown since it is a passive receiver. Hence, it depends only on the main channel information to choose the best relay.

The authors in [43] consider a different perspective, where they study the effect of hardware impairments on the secrecy performance of an HSTN. They propose an optimal relay selection scheme specifically for the case where the hardware of nodes is not ideal. Their results show the superiority of the proposed method over traditional round-robin scheduling for both perfect and impaired hardware cases. Moreover, they showed that in a high SNR situation, the secrecy capacity becomes lower in the case of impaired hardware, which increases the secrecy outage probability. However, by increasing the number of relays, the secrecy performance is enhanced. In [44], the authors study the secrecy performance of non-orthogonal multiple access (NOMA)-based users in the HSTN system. They consider the secrecy outage probability and define it as when a user fails to achieve the required level of secrecy. Their results showed that the channel condition of the weak user does not affect the secrecy performance.

Using a different type of relay, [45] proposed a 3D UAV relay HSTN system with an aerial eavesdropper around the UAV. The eavesdropper keeps track of the UAV relay to place himself at a suitable distance to obtain the transmitted data. They consider two scenarios where the eavesdropper is in a fixed place and a random location. They propose three strategies for UAV relay selection. The optimal method is based on the maximum-signal-to-noise ratio for the destination since the eavesdropper CSI is assumed to be unknown. However, they propose two less complex selection schemes due to the delay in channel estimation in satellite communications. One approach is based on the distance between the UAV relay and the destination and selecting the closest relay for transmission. The other approach randomly chooses the relay to assist in communication. Their results showed that selection based on SNR has the best performance, and the closest distance-based selection method outperforms the random selection. Furthermore, they found that the secrecy performance of the mobile UAV relaying outperforms the static conventional relaying. All discussed works assume no direct link between the satellite and the users due to shadowing and masking effects.

Table 3 summarizes the discussed work for PLS related to hybrid satellite-terrestrial networks. We compare the adversary model of all works and show that they consider passive adversary models using single or multiple eavesdroppers with a single antenna. Furthermore, all works use the wiretap channel's signal-to-noise ratio and channel state information to ensure security. However, these features can be challenging to acquire in the case of hidden adversaries. Similarly to the previous section, all work focuses on confidentiality requirements only without studying other security aspects, such as authentication of the relay or satellite.

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TABLE 3 Comparison between existing PLS works for hybrid satellite-terrestrial communication.

Work	Security goal	Relay type	Antenna	Adversary model	PLS technique	PLS feature	Security Measure
[37] (2018)	Confidentiality	Ground based DF	Single parabolic antenna	Multiple eavesdropper colluding and non-colluding	Relay technology	Signal to noise ratio	Security capacity
[38] (2018)	Confidentiality	Ground based DF	Single antenna	Single passive eavesdropper wiretap both links	Relay technology	Channel state information	Secrecy outage probability
[40] (2019)	Confidentiality	Ground basedDF, AF	Multiple antennas	Multiple eavesdropper colluding and non-colluding	Relay technology - multi-antenna	Signal to noise ratio	Secrecy outage probability
[43] (2019)	Confidentiality	Ground based DF	Single antenna with hardware impairments	Single eavesdropper wiretap relay link	Relay technology	Signal to noise ratio	Secrecy outage probability
[44] (2020)	Confidentiality	Ground based AF	single antenna	Single eavesdropper wiretap relay link	Relay technology	Signal to noise ratio	Secrecy outage probability positive secrecy capacity probability
[41] (2020)	Confidentiality	Aerial Interference relay	Multiple antenna	Near user eavesdropper wiretap satellite link	Relay technology - artificial Noise	Channel state information (instantaneous statistical)	Secrecy outage probability
[45] (2020)	Confidentiality	3D mobile UAV DF	Wide Beamwidth antenna	aerial eavesdropper in fixed & random positions	Relay technology	signal to noise ratio UAV to user distance	Secrecy outage probability non-zero secrecy capacity probability
[39] (2021)	Reliability and confidentiality	Ground-based DF	Single antenna-half duplex mode	Single eavesdropper wiretap satellite link	Relay technology - artificial Noise	Signal to interference plus-noise ratio	outage probability iIntercept probability
[42] (2022)	Confidentiality	Ground-based DF	single antenna	Single eavesdropper wiretap both link	Relay technology	Channel state information (known & unknown)	Secrecy outage probability

4.1.3 | Cognitive satellite terrestrial networks

The increase in demand for bandwidth due to new applications and the advancement of communication systems cause spectrum congestion in terrestrial and satellite networks. Currently, each network works on its respective spectrums, and congestion is a challenge for both networks. Cognitive satellite-terrestrial networks (CSTN) are based on the spectrum sharing between the two networks to solve the spectrum congestion issue by effectively using current frequency bands [46]. Cognitive radio is a promising paradigm and uses three primary schemes: underlay, overlay, and interweave. For satellite-terrestrial integration, underlay is widely considered to keep the interference of the terrestrial network with the satellite user under an acceptable level [47]. Although interference is a challenge in the cognitive network, it can be beneficial to enhance secure transmission if properly designed to degrade the wiretap channel more than the main channel. In this section, we review the current state-of-the-art PLS works in CSTN.

The authors of [48] proposed two beamforming techniques that utilize interference from the terrestrial network to improve the PLS security of the satellite network with joint optimization of transmit power and quality of service. Their network architecture is software-defined based cognitive satellite-terrestrial. A gateway is the control centre that manages the network with a database server that handles the system's CSI and spectral data. They consider single and multiple eavesdropper scenarios and use the penalty function approach to achieve the required beamforming weight vectors for the base station according to the data in the database server. The authors in [49] consider NOMA-based CSTN with decode and forward relay as a secondary network with hardware impairments and a passive eavesdropper. Specifically, they derive the closed-form expression of outage and inception probability with multiple antenna designs.

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Both works [50] and [51] propose a robust beamforming framework for CSTN with probabilistic quality of service constraints related to the user, eavesdropper, interference limit, and transmission power consumption. Both works use a terrestrial base station for beamforming to enhance the PLS. However, [50] used the terrestrial network as a source of friendly interference to secure the satellite user link in the presence of an eavesdropper. While in [51], the eavesdroppers wiretap terrestrial links instead of satellite links in simultaneous wireless information and power transfer (SWIPT) enabled network. Eavesdroppers are passive in both cases, with imperfect channel state information and angle estimates based on the second case. The primary goal of the beamforming technique in [50] is to minimize the transmit power while achieving at least minimum SINR to the satellite user, maximize the tolerance to SINR leakage to the eavesdropper, and constrain the interference power to the satellite user. Similarly, [51] paid more attention to maximizing the achievable secrecy rate in the worst case for all terrestrial users while achieving the required SINR and energy harvesting requirements for each user. In addition, they also constrain the power consumption of the base station and the interference limits of satellite stations. Due to these constraints, the problem becomes non-convex, so the authors use mathematical models to approximate the problem and then evaluate the computational complexity of their scheme and show its validity through numerical results.

Unlike the above techniques, [52] adds the assumption of an imperfect channel state for users and earth stations, not only eavesdroppers, and proposes a beamforming technique to exploit the base station as a source of green interference. Specifically, they designed a cooperative jamming scheme using a base station and cooperative terminal to degrade possible wiretap channels in the beamforming region, constraining the total power required for both terminals and SNIR and the interference limit for legitimate users. Ref. [53] uses a jamming-based technique to enhance the security of CSTN where a satellite receives the signal from a terrestrial user through RF links and then forwards it to another ground station using an optical link in the presence of an active eavesdropper for each link. They assume that one base station acts as a friendly jammer that uses pseudo-random sequences known only to legitimate users to generate artificial noise that affects illegitimate inception since they cannot decode it. The authors drive the closed form and asymptotic expressions of inception probability considering the presence and absence of the jammer.

The authors of [54] and [55] consider imperfect departure angles for the wiretap channel in a multibeam satellite system that shares a portion of the MMwave band with the cellular network to propose a secure beamforming scheme. Since beamforming techniques affect the system's power consumption, both works consider energy optimization in their design. In [54], they consider a wireless information-powered network (WIPT) and focus on the power minimization problem with constraints on SINR level for users and secrecy limit to energy receivers as potential eavesdroppers. However, in [55], they use a hybrid digital and analogue beamforming design to achieve a cost-performance trade-off and aim to maximize secrecy energy efficiency, which is the ratio between secrecy rate and consumed power. Both works use a cloud processing centre for resource allocation and control of the integrated network CSTN. Sequential convex approximation (SCA) was adopted to convert the non-convex problems into convex ones, and the results showed the advantage of the proposed schemes in different scenarios.

Table 4 summarizes the discussed PLS works related to CTSN. Similar to other satellite-terrestrial networks, the cognitive network works focus on the system's confidentiality and study the cases of single and multiple eavesdroppers. However, the energy efficiency constraint is vital for cognitive networks since most works use beamforming and artificial noise techniques that require additional energy and can affect system performance.

4.2 | Domain 2: satellite-based IoT networks

IoT communications are expected to provide massive connectivity in several services, including transportation, smart homes, manufacturing, health, agriculture, and maritime. These applications require extensive coverage in remote areas that terrestrial networks cannot support. In this case, the satellite network is a promising solution for providing IoT systems with a broad coverage, including harsh and rural areas. Specifically, LEO satellites can provide a trade-off between coverage and round trip latency around the earth, making them more favourable than MEO and GEO [56]. Moreover, hybrid satellite and terrestrial networks are proposed for maritime IoT to increase transmission efficiency and provide maritime service [57]. However, several challenges exist to achieve this connection, such as physical layer design and medium access mechanism. Security attacks are a severe concern in satellite IoT systems where IoT devices are considered vulnerable points exposed to different types of attacks such as interruption, spoofing, eavesdropping, and denial of service attacks [58]. For example, IoT devices can maliciously overwhelm one satellite by sending a large volume of packets, causing a denial of service attack and an interruption in the service. Applying intensive cryptographic techniques to satellite IoT devices is not applicable due to the power limitation of these devices and the short characteristics of their packets. In the following section, we discuss how physical layer security is used to secure satellite IoT communication through recent works.

In [59], the authors studied the physical layer secrecy analysis for satellite downlink signals with multiple mobile users and eavesdroppers in areas not covered by terrestrial networks. They proposed the FD-NOMA scheme that uses the frequency division of the downlink spectrum to provide efficient and secure access for multiple users using partial spectrum sharing between them. This scheme increases the inter-user interference, which PLS leverages to downgrade the eavesdropper's channel. At the same time, they propose a cooperative scheme that preserves the signal-to-noise ratio for legitimate users by interference cancellation. They derive the closed form of SINR for legitimate users and define a lower bound of secrecy rate for the system. In [60], they proposed a similar scheme to introduce artificial

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TABLE 4 Comparison between existing PLS works for cognitive satellite-terrestrial communication.

Work	Security goal	Network	Antenna	Adversary model	PLS technique	PLS feature	Security measure
[48] (2018)	Confidentiality	Software defined Network	Multibeam antenna	Single and multiple eavesdropper	Beamforming	Channel state information	Secrecy rate
[50] (2018)	Confidentiality	Downlink CSTN	Single antenna	Single eavesdropper wiretap satellite link	Beamforming	Probabilistic channel state information	SNIR constraints on user and eavesdropper
[49] (2020)	Confidentiality	NOMA based CSTN	Multi-antenna	Single eavesdropper wiretap terrestrial link	Relay technology	Hardware impairments	Secrecy outage probability Intercept probability
[51] (2020)	Confidentiality	mmWave CSTN with SWIPT	Parabolic antenna	Multiple eavesdroppers Wiretap terrestrial link	Beamforming	Angle-based imperfect channel state information	Achievable secrecy rate
[52] (2021)	Confidentiality	mmWave CSTN	Multiple antenna	Multiple eavesdroppers wiretap terrestrial link	Beamforming and artificial noise	Imperfect channel state information	Signal to noise ratio
[53] (2021)	Confidentiality	Underlay CSTN	—	Two eavesdroppers wiretap satellite links	Artificial noise	Pseudo-random sequences	Inception probability
[54] (2021)	Confidentiality	WIPT cloud processing architecture	Multibeam antenna	energy receivers	Beamforming	Angle-of-departure based channel state information	Achievable secrecy rate
[55] (2021)	Confidentiality	mmWave CTSN cloud processing architecture	Array fed reflector antenna	Multiple eavesdropper wiretap terrestrial link	Hybrid beamforming	Angle-of-departure based channel state information	Secrecy energy efficiency

interference and leverage it to affect the eavesdropper channel, using overlapping pulse shapes. They analyse the mutual information and secrecy capacity if the eavesdropper cannot resolve the time packing interference and in case he has an estimation module, and in both cases, security capacity is achieved.

Ref. [61] consider cognitive satellite-terrestrial network for IoT services with a UAV malicious eavesdropper. They use terrestrial base stations as beamformers and friendly jammers utilizing inter-segment interference to reduce the quality of the eavesdropper channel in a 3D wiretap environment. The proposed beamforming scheme depends on spatial correlation with energy consumption and secrecy rate constraints to maximize secrecy energy efficiency. On the other hand, authors in [62] assume satellite-UAV cognitive network architecture, where each network has its users, and share the spectrum divided into different channels, each channel serving the satellite user and UAV user. They consider terrestrial eavesdroppers wiretap the signals from UAVs and aim to maximize the secrecy rate of UAV users by optimizing power, resource allocation, and interference constraints. They compare their scheme with the other two power allocation schemes and show the superiority of the proposed method in providing a higher secrecy rate.

Ref. [63] proposed a user scheduling scheme for multiuser satellite and wireless sensors network to enhance secrecy performance. They derive the closed-form expression of secrecy outage probability and average secrecy capacity when the CSI of the eavesdroppers is unknown. They assume a single antenna for all users and eavesdroppers and utilize time division multiple access to ensure that one user is using the service at each time. Similarly, the authors in [64] consider a multi-beam satellite that enables vehicular communications with UAVs as amplify and forward relays to enhance the main channel between the satellite and vehicles where rain attenuation is assumed for the satellite channel. Moreover, UAVs act as a jammer to produce artificial noise that affects the eavesdropper channel. They jointly optimize the power for satellite beamforming and power allocation for UAV relays to maximize the secrecy rate of satellite-to-vehicle links without affecting the QoS requirements. They optimize the non-convex optimization problem using several methods and compare the secrecy rate for the proposed technique with a case without artificial noise and a benchmark case without UAV cooperation to showcase the improvement in security performance resulting from UAV assistance.

Table 5 summarized the discussed works related to PLS on satellite-based IoT networks, which mainly focus on system confidentiality with a single eavesdropper and consider the security of downlinks only. Authenticating satellite and IoT devices using PLS to prevent spoofing and impersonation attacks has

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TABLE 5 Comparison between existing PLS works for IoT networks.

Work	Security goal	Network	Antenna	Link	Adversary model	PLS technique	PLS feature	Security Measure
[59] (2019)	Confidentiality	Multi-user satellite	Multi-antenna	Downlink	Single eavesdropper	Information theory	Inter-user interference	Secrecy rate
[60] (2019)	Confidentiality		Multi-antenna	Uplink & Downlink	Single eavesdropper	Information theory	Time pulses	Secrecy Capacity
[61] (2020)	Confidentiality	Cognitive satellite terrestrial network	Multi-antenna	Downlink	Malicious UAV	Beamforming	Spatial correlations	Secrecy energy efficiency
[63] (2020)	Confidentiality	Multi-user satellite	Single antenna	Downlink	Colluded passive eavesdropper	Information theory	Time division multiple access	Secrecy outage probability average secrecy capacity
[62] (2021)	Confidentiality	Cognitive satellite and UAV	Single antenna	Downlink	Terrestrial eavesdropper	Information theory	Large-scale CSI	Secrecy rate
[64] (2022)	Confidentiality	Multi-beam Satellite	Multi-antenna	Downlink	Terrestrial eavesdropper for each beam	Relaying & Artificial noise	Perfect CSI	Secrecy rate

not been studied yet. Moreover, protection against multiple malicious IoT devices and denial of service attacks (common in the case of IoT devices) are not available in the literature. Furthermore, more attention should be paid to studying the security of uplink communication, which is critical for IoT systems, as IoT devices transfer a massive amount of data to the satellite due to their increasing number, and these data can be eavesdropped or altered if no proper uplink security mechanism is applied.

4.3 | Domain 3: navigation satellite systems

Global navigation satellite systems (GNSS) are localization systems that use satellite signals to share information about location, speed, and time with Earth users. GPS [65] and GALILEO [66] are examples of GNSS widely used in several applications, such as intelligently connected vehicle networks, mobile applications, and maritime. However, GNSS is vulnerable to spoofing attacks because of its signal's lack of authentication mechanism. Specifically, publicly known spreading codes allow attackers to forge messages and send them to users as legitimate messages, causing incorrect positioning, which can lead to severe consequences [67]. The attacker can jam legitimate messages to prevent their reception by legitimate receivers and give more chances to illegitimate ones. Through the years, several mechanisms have been proposed to prevent GNSS spoofing attacks. In this section, we focus on physical layer schemes used to detect and mitigate GNSS spoofing.

Due to the invisibility of changing the algorithms of GNSS, the proposed methods are superposition-based. In [68], the proposed scheme uses an authentication message and artificial noise on top of the GNSS message to authenticate it. In this paper, they consider three channels in the system: 1) the navigation channel, which is the broadcast signal from the satellite to all users. 2) the attack channel, where the eavesdropper transmits the spoofing signal to the legitimate user. 3) the authenticated channel where a ground segment can communicate with all users through infinite bandwidth, and the eavesdropper has no control over this channel (authenticated through higher layer protocol). The legitimate user uses the authenticated channel to know the signature and artificial noise after revealing it by the system and check if the received navigation signal coordinates are included. They assume that the attacker receives the signal from the authenticated channel but does not have control over it. Two attack models were considered in the analysis: the generation attack, where the attacker generates a simple navigation signal and uses it, and the replay attack, where the adversary combines the signal with a previously used authentication signal. Although this method utilizes PLS, it relies on the upper layer algorithms to secure the authenticated channel, which is essential for the design of the system. On the other hand, the authors in [69] use satellite signals' radio frequency spatial characteristics to identify spoofing signals. Using the K-means cluster algorithm, they use consultation figures to extract radio features and fingerprint satellite signals. Specifically, the clustering algorithm generates multiple cluster centres according to the signal features, which helps to detect spoofing by analysing the centre change. They verify the validity of the proposed scheme using experimental analysis through accurate satellite signals and spoofed signals transmitted by the Universal Software Radio Platform.

Similarly, the authors in [70] use device fingerprinting to create signatures for GPS satellite transmissions and allow identifying the spoofed and authentic signals. The attack model considered in this paper is similar to [68], where spoofer attackers can generate genuine GPS messages or use relay messages to spoof legitimate receivers and induce wrong position, velocity, and time messages. They propose a feature extraction method for fingerprinting based on the internal hardware architecture of GPS receivers. Specifically, they use early–late phase (ELP) correlator outputs, code-lock-detector (CN0), carrier-lock-detector (CLT) for feature extraction, and multivariate normal distribution (MVN) as a scoring metric to measure similarity. To find thresholds for genuine GPS signals, they first train the feature

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extraction method using a sample spoofed-free dataset and then test it on spoofed signals to calculate the MVN for each observation, and low MVN means spoofed signal. These thresholds are used offline for real-time spoofing detection after the cross-validation phase. They collect live datasets and perform an experimental analysis for the proposed algorithm under different attack scenarios, including location, time, and multipath.

Depending on feature analysis, the authors in [71] propose a spoofing detection method using a support vector machine (SVM) to analyse signal quality monitoring (SQM), moving variance (MV), improved SQM moving average (MA), early–late phase, carrier-to-noise ratio–MV, and clock offset rate. After feature extraction, the data is preprocessed using normalization techniques and divided into train and test datasets. They perform offline training and testing phases for SVM using the Texas Spoofing Test Battery (TEXBAT) dataset under different kernel and activation functions to get the best performance. They reach 92.31% accuracy compared to other detection methods and provide other indices evaluation such as recall, precision, and F1.

Another detection approach is proposed in [72], where the iridium satellite constellation is utilized to detect GNSS spoofing using unencrypted IRIDIUM Ring Alert (IRA) messages that can be received worldwide from all Iridium satellites. In particular, using thousands of messages from the data collection campaign, they reverse engineer satellite parameters such as speed, packet interval time, and satellite coverage. They assume a system model suitable for remote areas such as deserts and oceans where the receiver cannot check their location with other sources. Detecting spoofing depends on collecting IRA messages on the receiver side and compensating for the movement. The receiver then estimates the location using the longitude and latitude of the collected beams. Finally, the estimated position is compared to the position received by the GNSS satellite, and if it is greater than the specific threshold, the signal is considered malicious.

Ref. [73] consider another application scenario where they propose a spoofing detection method for electric substations since it relies on GNSS services for time transfer. The method utilizes multiple antenna receivers for GPS signals in proximity where the spoofer cannot send a spoofing signal for each antenna, so only a limited number of antennas will give an incorrect location. Each antenna is connected to a GNSS clock used in a conventional substation that estimates the position using the received signal and then sends it to the detector. The detector will compare all acquired positions, decide the expected location, and raise the alarm if a spoofing attack is detected. They validated that their method can detect the spoofing attack in 4 to 10 s, which eliminates the attack effect on the substation.

Table 6 summarizes the work discussed related to PLS in navigation satellite systems, which clearly focuses on PLS authentication and spoofing detection mechanisms since navigation satellite systems are vulnerable to spoofing and impersonation attacks. The confidentiality of these systems is not considered critical since the messages are not secret (they only provide location and time information). However, the integrity of these messages should be preserved to prevent malicious alteration during transmission. In addition, navigation satellite services are vital to many applications; therefore, more attention should be paid to studying the availability of the system.

4.4 | Domain 4: free space optical satellite communication

Optical wireless communication (OWC) uses optical carriers such as visible light and Infrared to transmit information. Free space optical (FSO) is a type of OWC that uses directive laser beams to transmit data wirelessly in unguided free space media to provide high-speed communication between two points. The optical medium is an unlicensed spectrum that overcomes the limitations and congestion of the current radio frequency band and suffers from less electromagnetic interference [74]. Furthermore, FSO can provide a high data rate with low latency for several kilometres without expensive infrastructure, reducing installation and maintenance costs. Satellite communications, including uplink, downlink, and interlinks, start incorporating FSO links to leverage its advanced features and improvements. Specifically, FSO allows higher data rates to reach gigabits per second (Gbps), which is significantly better than the RF band. Moreover, FSO has inherent security features that it cannot penetrate walls and can support quantum cryptography when fibre optic infrastructure is unavailable [75].

On the other hand, FSO suffers from fading and attenuation induced by atmospheric turbulence such as fog, rain, and snow [76]. In addition, beam divergence caused by beam wanders causes misalignment between the transmitter and the receiver. These conditions can affect the link's reliability and availability and limit its capacity to short-distance ranges. To address these shortcomings in FSO and take advantage of its features simultaneously, a hybrid Satcom system between RF and FSO is designed. RF links are more robust to weather conditions and mature than FSO development. Several hybrid Satcom designs have been proposed in the literature, such as hard switching [77], adaptive switching [78], and HAPS relay-based design. In this section, we discuss the physical layer security analysis available in the literature for FSO and a hybrid system that uses both RF and FSO.

The authors in [79] and [80] design an FSO satellite system with ground optical stations. In [79], the ground stations send N users information by optical feeder link to the satellite, which acts as a relay that decodes the optical wave received and sends it as beams to multiple earth users with earth eavesdropper for both links. However, the authors in [80] consider the uplink connection between the ground station and a GEO satellite with a nano-satellite eavesdropper. They propose two PLS secrecy coding protocols: one-way and two-way protocols with two types of photodetectors: threshold and PNR. Their two-way scheme ensures secrecy even when the eavesdropper's channel is better than the legitimate channel. Both works analyse the secrecy capacity using the SNR between the legitimate user and eavesdropper. In [79], the authors consider two scenarios, ZF technique and without ZF technique, for the analysis. Without the ZF technique, the satellite decodes the received data and

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TABLE 6 Comparison between existing PLS works for navigation satellite system.

Work	Security goal	Satellite	Adversary model	PLS technique	PLS feature	Security measure
[68] (2018)	Authentication	Galilo	Generation attack - replay attack	Artificial noise	Random generated codes	Channel gain - channel estimation error
[69] (2018)	Spoofing detection	GPS	—	Hardware fingerprinting	Spatial distribution characteristics	Euclidean distance
[70] (2020)	Spoofing detection	GPS	Generation attack replay attack	Hardware fingerprinting	Early late phase code-lock-detector carrier-lock-detector	Equal error rates
[72] (2020)	Spoofing detection	GNSS	Generate fake messages using SDR	Parameters reverse engineering	IRA iridium messages	False positive rate
[71] (2021)	Spoofing detection	GPS	Generate spoofing signal using user tracking signal	Feature extraction	Quality monitoring (SQM) moving variance (MV) improved SQM moving average (MA) early–late phase carrier-to-noise ratio MV and clock offset rate	Detection accuracy
[73] (2022)	Spoofing detection	GNSS	Generate spoofing signals for electric substation	Multiple antenna	Spatial area	Detection speed

directly forwards it to the users, while with ZF, the satellite codes the beams using the ZF technique before sending. The results showed that the ZF technique improves the security capacity with specific positions.

In [81], the authors consider a hybrid system RF link between the satellite and the ground station (the long distance) and an FSO link between the station and destination (short distance). The station is the decode and forward relay between source and destination, and the eavesdropper is assumed to be on the earth trying to get access to the data sent by satellite. They analyse the average secrecy capacity and secrecy outage probability under different satellite and FSO channel conditions, relaying schemes, and FSO detection methods. RF link conditions have more impact on the system secrecy than the FSO, while secrecy diversity depends on FSO conditions when using amplify and forward relaying methods.

Both [82] and [83] use a high-altitude platform station (HAPS) as a relay between a LEO satellite and the terrestrial destination. The HAPS has an FSO downlink with the satellite and an RF downlink with the Earth station. The eavesdropper in this system is passive and located on the earth's surface. They derive the secrecy outage probability (SOP) and the probability of positive secrecy capacity of the system and analyse them under different channel conditions. The results in [82] showed that the RF link has more impact on the system performance. At the same time, both analyses emphasize the criticality of pointing errors and channel shadowing on the system's security.

Table 7 summarizes the works related to PLS on FSO satellite communication, which shows that the use of PLS in optical satellite communication is still limited as most works consider a hybrid system of optical and radio frequencies and relay techniques between them. Therefore, the PLS technique needs to be investigated more in FSO links with a focus on how distance and weather conditions can affect it.

4.5 | Domain 5: inter-satellite communication

Inter-satellite communication links connect satellites to each other. Typically, satellites are deployed in groups where communication between them is essential for service continuity. Satellites must share their navigational and mobility data to keep the formation in the correct shape [84]. The connection between satellites increases the network throughput, where the message is routed through satellites until it reaches its destination without the need for ground stations. Specifically, this communication creates a network in the space that improves the transmission and allows service handover. However, intersatellite links are complex, since they connect satellites from various levels (GEO, MEO, and LEO), with different speeds and altitudes that can affect the link availability and quality. These links can be in radio frequency or optical communication.

The complexity of inter-satellite links increases its exposure to adversaries from multiple levels, such as satellites from other constellations, spacecraft, and adversaries on Earth. In addition, inter-satellite links have different channel states than ground links because of the high mobility of the satellites and fewer fading sources. Despite the complexity and vulnerability of the inter-satellite links, security mechanisms to protect the space networks are still limited. Inter-satellite links are vulnerable to attacks that compromise confidentiality and authentication protocols where the satellites are not identified before initiating a connection, and data are not protected against any adversary in the middle of the communication. Furthermore, adversaries can

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TABLE 7 Comparison between existing PLS works for FSO satellite communication.

Work	Security goal	Link type	Relaying scheme	Channels types	Security measurement
[81] (2019)	Confidentiality	Downlink	Amplify and forward	Shadowed-Rician For RF	Average secrecy capacity
			Decode and forward	GammaGamma for optical	Secrecy outage probability
[79] (2020)	Confidentiality	Downlink and uplink	Decode and forward	GammaGamma for optical	Secrecy capacity
[80] (2020)	Confidentiality	Uplink	—	Poisson channel	Secrecy capacity
[82] (2021)	Confidentiality	Downlink	Decode and forward	Shadowed-Rician For RF	Secrecy outage probability
				GammaGamma for optical	Probability of positive secrecy capacity
[83] (2022)	Confidentiality	Downlink	Amplify and forward	Shadowed-Rician For RF	Secrecy outage probability
				GammaGamma for optical	Probability of positive secrecy capacity

target the availability of inter-satellite links to cause service interruption by overwhelming the link using a compromised satellite or malicious spacecraft.

Some authors apply the PLS framework to inter-satellite links such as [85, 86]. In [85], the authors use Doppler frequency shift to generate keys for securing the inter-spacecraft links. In [86], the authors use the Doppler frequency shift as a physical layer authentication method for satellite identification, utilizing satellite voting. The Doppler frequency shift is the change in frequency relative to the motion. Both works use radio frequency links and satellite mobility information, such as speed and locations, to estimate the Doppler frequency shift of each satellite. Since the mobility information of each satellite is already shared on the network, this method does not introduce overhead or additional channels.

In [85], the authors assume communication between two spacecraft, a transmitter and a receiver, in deep space using a time division duplexing (TDD) system. Both legitimate users will experience identical Doppler frequency shifts using their high mobility. At the same time, the eavesdropper observes a different shift since her relative velocity and position vary from the transmitter and receiver, which allows them to use the Doppler frequency shift as a source of secret keys. The secret key generation process starts with pilot transmission and reception between the transmitter and receiver, assuming that they have the same velocity at this step. Then, each spacecraft will estimate the nominal power spectral density for each communication block separately, which measures the power of the content of the message versus its frequency. After that, they will quantize the achieved value to obtain the raw secret sequence. The authors perform a Monte Carlo simulation for the proposed scheme and analyse several aspects, such as the key disagreement rate and maximum achievable key rate, to prove the practicality and robustness of the technique.

In [86], the authors consider a set of legitimate LEO satellites and one satellite that needs to be authenticated. When satellites receive a message from a suspect satellite, they will calculate the Doppler frequency using nominal power spectral density estimation, which measures the power of the content of the message versus its frequency. All satellites will compare the resulting Doppler frequency with the expected one for the transmitter, which can be calculated by any satellite, since the transmitter's location and speed are known. Then, each satellite will decide if this transmitting satellite is legitimate based on a specific threshold and send this decision to the fusion centre. In the fusion centre, the algorithm will collect all decisions and use the decision method to make the final judgment on the satellite's identity. The authors analyse the performance of the algorithm with different numbers of satellites and lengths for the time slot. They also compare three fusion decision methods, OR, AND, and majority rule, and measure their spoofing detection and false alarm probabilities. The results showed that the majority rule outperforms others, achieving a high spoofing detection rate with low false alarm probability.

PLS in optical inter-satellite links is still an uncovered area in the literature due to its intrinsic security features earned by the inability to penetrate through walls. However, optical links are still vulnerable to attacks such as eavesdropping, as the authors in [87] propose eavesdropping techniques for space networks where HAPS is the eavesdropper for LEO satellite communication.

Table 8 summarizes recent work related to PLS for inter-satellite communication. Although inter-satellite communication is critical to the satellite system, its security is not yet well studied in the literature. Both radio and optical inter-satellite links are vulnerable to multiple attacks, such as eavesdropping, spoofing, and jamming.

In LEO satellite constellations, the situation is exacerbated, where attacking these inter-satellite links will often affect significant services, especially those requiring the transfer of satellite-to-satellite messages.

Moreover, inter-satellite links exhibit unique features that differ from satellite-to-earth links, which prevent the usage of the available PLS techniques in the satellite-to-earth setting. Therefore, more research is required to study how these features can be used with PLS to provide security against possible attacks.

5 | RESEARCH GAPS AND FUTURE DIRECTIONS

Physical layer security in satellite systems still has many open issues that need to be investigated. In this section, we

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TABLE 8 Comparison between existing PLS works for Inter-satellite communication.

Work	Link	Security goal	Antenna	PLS feature	Security measure
[85] (2021)	Radio frequency inter-spacecraft link	Secret key generation	Omni-directional antenna	Doppler frequency shift	Maximum achievable key Rate Key disagreement rate
[86] (2022)	Radio frequency inter-satellite link	Authentication	Single omnidirectional antenna	Doppler frequency shift	Spoofing detection rate False alarm probability

highlight open research opportunities and future directions that we identify through our survey study.

5.1 | Availability schemes

Due to the importance of satellite systems in the next emerging network architecture, ensuring their availability is essential to avoid system outages or interruptions. Satellite systems can suffer from denial-of-service attacks if the satellite or link is overwhelmed with malicious packets that the satellite cannot handle and cause service outages for legitimate users. Moreover, anti-jamming techniques for satellite communications need to be investigated since adversaries can target wireless signals through multiple types of jamming, causing transformations to the signals that prevent correct decoding and processing. PLS techniques are promising in anti-jamming methods and cooperative jamming to prevent various attacks in 6G networks [88]. Therefore, more work should consider PLS availability techniques in satellite scenarios to study its feasibility with the unique physical characteristics of satellite communication.

5.2 | Uplink secrecy techniques

The current literature focuses mainly on securing downlink communication from the satellite to other receivers. However, uplink communication from any terrestrial or aerial device to the satellite is vulnerable to attacks such as spoofing and alteration by malicious adversaries. Especially in broadband satellite communication, the uplink handles a huge amount of data sent by users to the satellite, and these data must be secured through schemes suitable for heterogeneous devices and scenarios. Moreover, uplink communication for the satellite can be used to overwhelm the satellite using multiple techniques of denial-of-service attack. Therefore, researchers should focus more on securing uplink communication using PLS, considering performance and power constraints.

5.3 | Smart threat models

Most current literature assumes that the eavesdropper is passive, with limited resources and a single antenna, and does not actively affect the main channel. They do not consider the case where eavesdroppers can extract CSI of the main channel by active interception, reducing the effectiveness of the current schemes. These assumptions are becoming ideal with advancements in adversaries' resources, such as the number of antenna and signal processing techniques. Moreover, the widespread assumption of static and non-moving eavesdroppers can be enhanced by having dynamic eavesdroppers with unknown locations, such as UAVs. Therefore, more complex adversary models should be considered in satellite systems to cooperate with advanced eavesdropping capabilities and design more robust solutions.

5.4 | Authentication and anti-spoofing techniques

Through analysis of the available schemes for authenticating satellite signals, we notice that most techniques are focused on GNSS satellites due to their significance and vulnerability to spoofing. However, other types of satellites used in different services, such as broadband and IoT connection, are vulnerable to spoofing or impersonation attacks that affect their service and give access to unauthorized devices. In particular, failing to authenticate the messages sent to the satellite can cause several consequences due to processing malicious messages/code received. In wireless systems such as satellites, the adversary sends signals with a higher power than the legitimate user to deceive the receiver into considering his spoofed message as the legitimate one [89]. Hence, the adversary will get access to the service and information intended for the legitimate user. As a result, anti-spoofing PLS strategies need investigation in satellite scenarios for different services to prevent unauthenticated access or information leakage.

5.5 | Integrity techniques

Message manipulation detection techniques using PLS are not yet considered in the literature for any wireless communication. Integrity violation attacks such as man-in-the-middle intercept the message during transmission, modify or manipulate it, and re-transmit it. This change in the message must be detected by the receiver to prevent the transmission and processing of incorrect information. Satellite systems are vulnerable to these attacks, where adversaries can intercept the signal sent by satellite to the terrestrial user or vice versa and manipulate it either to disturb the service or to get access by exploiting these messages. PLS techniques to achieve integrity need to be investigated using different wireless signals, including satellite systems.

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5.6 | PLS for inter-satellite link communication

Inter-satellite communication is essential for providing high-speed communication and increasing network throughput by connecting satellites to each other. With the increasing interest in LEO satellites that are relatively near the Earth and advancements in adversary techniques, attacks on inter-satellite communication have become feasible. There is limited focus on securing inter-satellite communications in the literature for both RF and optical links. In particular, inter-satellite links have unique physical layer characteristics in terms of fading, Doppler shift, and weather condition effects that can be exploited to secure communication. Moreover, optical inter-satellite links provide high speed and bandwidth, but they are still vulnerable to attacks such as eavesdropping, and there are no proposed solutions yet in the literature. PLS strategies are promising techniques that must be studied extensively in inter-satellite communication scenarios by investigating their physical features and possible attacks.

5.7 | Machine learning based PLS

Wireless networks are becoming more complex with heterogeneous devices and high mobility requirements. Recently, machine learning techniques have emerged to support PLS in handling network complexity by intelligently handling signal processing and feature exploitation. Several supervised learning and unsupervised ML techniques are used for intelligent PLS in different categories, such as physical layer authentication, antenna selection, and relay node selection [90]. The proposed ML-based PLS schemes consider various wireless communications, including device-to-device, cognitive networks, and NOMA. However, ML-based algorithms are not explored in satellite systems scenarios, and there is no analysis of the applicability of the proposed methods in the satellite case. Therefore, more research is required to benefit from ML techniques in simplifying PLS in satellite systems and consider its unique features.

6 | CONCLUSION

Satellite communication systems are vulnerable to attacks due to their broadcast nature and the heterogeneity of connected devices. PLS schemes are lightweight security approaches that exploit physical layer characteristics to provide the required security while ensuring minimal overhead, making them suitable for several communication types, including satellite systems.

In this paper, we first classify contemporary satellite communication networks into five domains and then analyse the use of PLS in each domain; these domains are hybrid satellite-terrestrial networks, satellite-based IoT, navigation satellite systems, FSO-based satellites, and inter-satellite communications. After providing the necessary background about satellite networks and PLS techniques, we reviewed the state-of-the-art PLS solutions in each domain. We compare their approaches, adversary models, and results. At the end of our survey, we highlighted a few open research gaps for PLS in satellite systems and proposed a few potential future directions, which we believe the community must focus on. These identified gaps include utilizing PLS techniques to provide critical security goals such as availability, authentication, and integrity to satellite network links, including uplink communication and inter-satellite links. Moreover, we identified the need to improve the threat models used in PLS scenarios to cover more advanced adversary capabilities and design more robust solutions. Finally, we highlight the opportunity to leverage machine learning techniques in simplifying PLS systems for satellite network scenarios.

AUTHOR CONTRIBUTIONS

Nora Abdelsalam: Conceptualization; data curation; investigation; methodology; writing—original draft. **Saif Al-Kuwari**: Funding acquisition; supervision; validation; visualization; writing—review and editing. **Aiman Erbad**: Visualization; writing—review and editing.

ACKNOWLEDGEMENTS

This publication was made possible by GSRA grant # GSRA8-L-2-0521-21046 from the Qatar National Research Fund (a member of Qatar Foundation) and a TÜİTAK-QNRF Joint funded grant # AICC03-0530-200033. The findings herein reflect the work and are solely the responsibility of the authors.

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CONFLICT OF INTEREST STATEMENT

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

Data sharing not applicable to this article as no datasets were generated or analysed during the current study.

ORCID

Nora Abdelsalam https://orcid.org/0000-0002-4393-9507

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How to cite this article: Abdelsalam, N., Al-Kuwari, S., Erbad, A.: Physical layer security in satellite communication: State-of-the-art and open problems. IET Commun. 19, e12830 (2025). https://doi.org/10.1049/cmu2.12830

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